

1 Summary

Model (Run-name): model3 (test)

Iteration: 9

TAC: 63763.52533345599

Took: 171.75772094726562s

Total Time: 462.83159828186035s

Epsilons:

- ε_{LMTD} : 1e-06
- $\varepsilon_{Balancing}$: 1e-06
- ε_{Area} : 1e-06
- ε_{Beta} : 1e-06

Gurobi Tolerances:

- IntFeasTol: 1e-05
- FeasibilityTol: 1e-06
- OptimalityTol: 1e-06

Gamsworld Solution: http://www.gamsworld.org/minlp/minlplib2/html/heatexch_gen3.html

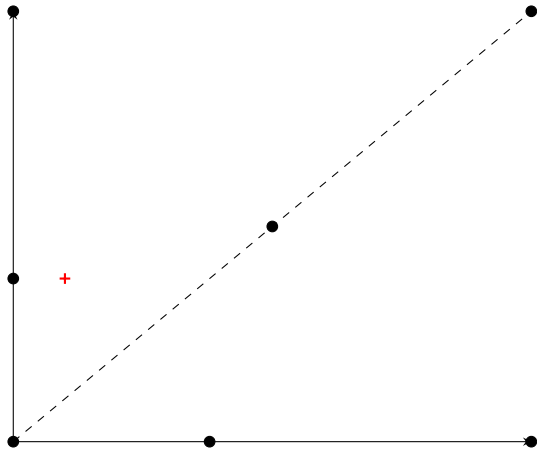
2 Results

Variable	Value	Variable	Value	Variable	Value
$z_{2,1,2}$	1.0	$dt_{3,5,2}$	22.3	$q_{3,5,2}$	1544.2900000000004
$z_{2,2,1}$	1.0	$dt_{3,5,3}$	13.545945945951658	$q_{4,3,2}$	1539.7200000000003
$z_{3,1,2}$	1.0	$dt_{4,3,2}$	50.0	$q_{5,4,2}$	1634.85
$z_{3,5,2}$	1.0	$dt_{4,3,3}$	111.1	$q_{cu,1}$	586.9599999999999
$z_{4,3,2}$	1.0	$dt_{5,4,2}$	22.2	$q_{cu,3}$	610.4400000000085
$z_{5,4,2}$	1.0	$dt_{5,4,3}$	24.3355932203388	$q_{cu,5}$	724.5599999999972
$z_{cu,1}$	1.0	$dt_{cu,1}$	120.0	$t_{1,1}^{(H)}$	160.0
$z_{cu,3}$	1.0	$dt_{cu,3}$	66.84594594595168	$t_{1,2}^{(H)}$	160.0
$z_{cu,5}$	1.0	$dt_{cu,5}$	66.53559322033882	$t_{1,3}^{(H)}$	160.0
$A_{2,1,2}^{\beta}$	2.9471014397878545	$f_{2,1,2}^H$	10.6	$t_{2,1}^{(H)}$	248.9
$A_{2,2,1}^{\beta}$	5.378842799033671	$f_{2,2,1}^H$	5.8254725472547255	$t_{2,2}^{(H)}$	187.84245283018686
$A_{3,1,2}^{\beta}$	0.5680195990179859	$f_{3,1,2}^H$	1.4248913718187304	$t_{2,3}^{(H)}$	137.8
$A_{3,5,2}^{\beta}$	15.525824163083303	$f_{3,5,2}^H$	9.58590937306018	$t_{3,1}^{(H)}$	226.7
$A_{4,3,2}^{\beta}$	6.675812719571878	$f_{4,3,2}^H$	12.6	$t_{3,2}^{(H)}$	226.7
$A_{5,4,2}^{\beta}$	13.945813823426127	$f_{5,4,2}^H$	12.26444111027757	$t_{3,3}^{(H)}$	106.84594594595168
$A_{cu,1}^{\beta}$	3.357327282820032	$f_{2,1,2}^C$	5.3045	$t_{4,1}^{(H)}$	271.1
$A_{cu,3}^{\beta}$	4.796859386349519	$f_{2,2,1}^C$	6.1	$t_{4,2}^{(H)}$	271.1
$A_{cu,5}^{\beta}$	5.324435317424491	$f_{3,1,2}^C$	2.2954999999999997	$t_{4,3}^{(H)}$	148.9
$A_{2,1,2}$	9.363566933860874	$f_{3,5,2}^C$	13.9	$t_{5,1}^{(H)}$	198.9
$A_{2,2,1}$	16.512573095835464	$f_{4,3,2}^C$	8.4	$t_{5,2}^{(H)}$	198.9
$A_{3,1,2}$	1.6200289353893638	$f_{5,4,2}^C$	17.3	$t_{5,3}^{(H)}$	106.53559322033882
$A_{3,5,2}$	103.45829612760171	$RecLMTD_{2,1,2}$	0.015004301807488477	$t_{1,1}^{(C)}$	160.0
$A_{4,3,2}$	23.668654734065058	$RecLMTD_{2,2,1}$	0.021686449732637232	$t_{1,2}^{(C)}$	160.0
$A_{5,4,2}$	82.7196877037192	$RecLMTD_{3,1,2}$	0.017796893251953626	$t_{1,3}^{(C)}$	60.0
$A_{cu,1}$	7.5275930538960925	$RecLMTD_{3,5,2}$	0.05694497258187349	$t_{2,1}^{(C)}$	221.7
$A_{cu,3}$	13.643579101040302	$RecLMTD_{4,3,2}$	0.013066243553344306	$t_{2,2}^{(C)}$	115.6
$A_{cu,5}$	16.235135805353295	$RecLMTD_{5,4,2}$	0.0430080646837088	$t_{2,3}^{(C)}$	115.6
$dt_{2,1,2}$	27.842452830189046	$RecLMTD_{cu,1}$	0.010901005342462128	$t_{3,1}^{(C)}$	221.1
$dt_{2,1,3}$	77.8	$RecLMTD_{cu,3}$	0.018997841288054415	$t_{3,2}^{(C)}$	221.1
$dt_{2,2,1}$	27.2	$RecLMTD_{cu,5}$	0.019045856015444104	$t_{3,3}^{(C)}$	37.8
$dt_{2,2,2}$	72.24245283018871	$q_{2,1,2}$	530.45	$t_{4,1}^{(C)}$	176.7
$dt_{3,1,2}$	66.7	$q_{2,2,1}$	647.2099999999999	$t_{4,2}^{(C)}$	176.7
$dt_{3,1,3}$	46.845945945951684	$q_{3,1,2}$	229.55	$t_{4,3}^{(C)}$	82.2

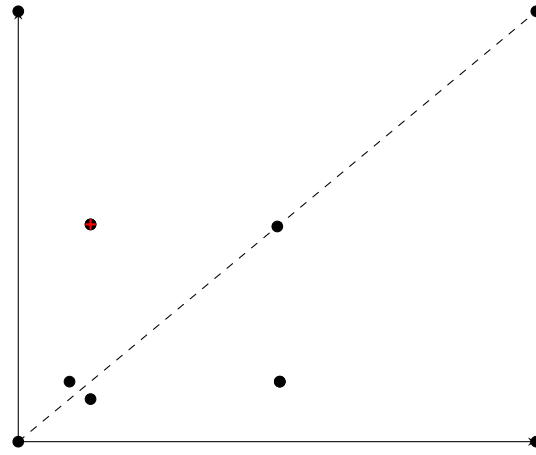
Variable	Value	Variable	Value
$t_{5,1}^{(C)}$	204.4	$b_{2,1,2}^{C,out}$	848.72
$t_{5,2}^{(C)}$	204.4	$b_{2,2,1}^{C,out}$	1352.37
$t_{5,3}^{(C)}$	93.3	$b_{3,1,2}^{C,out}$	367.2799999999975
$t_{2,1,2}^H$	137.8	$b_{3,5,2}^{C,out}$	2841.1600000000003
$t_{2,2,1}^H$	147.95284246475143	$b_{4,3,2}^{C,out}$	1857.2400000000002
$t_{3,1,2}^H$	65.6	$b_{5,4,2}^{C,out}$	3056.91
$t_{3,5,2}^H$	65.6		
$t_{4,3,2}^H$	148.9		
$t_{5,4,2}^H$	65.6		
$t_{2,1,2}^C$	150.87721563365653		
$t_{2,2,1}^C$	221.7		
$t_{3,1,2}^C$	90.20394736842071		
$t_{3,5,2}^C$	204.4		
$t_{4,3,2}^C$	221.1		
$t_{5,4,2}^C$	176.7		
$b_{2,1,2}^{H,in}$	1991.1300000000538		
$b_{2,2,1}^{H,in}$	1449.9601170117012		
$b_{3,1,2}^{H,in}$	323.02287399130614		
$b_{3,5,2}^{H,in}$	2173.125654872748		
$b_{4,3,2}^{H,in}$	3415.86		
$b_{5,4,2}^{H,in}$	2439.3973368342085		
$b_{2,1,2}^{H,out}$	1460.6800000000537		
$b_{2,2,1}^{H,out}$	802.7501170117011		
$b_{3,1,2}^{H,out}$	93.4728739913087		
$b_{3,5,2}^{H,out}$	628.8356548727477		
$b_{4,3,2}^{H,out}$	1876.14		
$b_{5,4,2}^{H,out}$	804.5473368342084		
$b_{2,1,2}^{C,in}$	318.27		
$b_{2,2,1}^{C,in}$	705.16		
$b_{3,1,2}^{C,in}$	137.73		
$b_{3,5,2}^{C,in}$	1296.87		
$b_{4,3,2}^{C,in}$	317.52		
$b_{5,4,2}^{C,in}$	1422.0600000000002		

3 *RecLMTD* Tangent Points

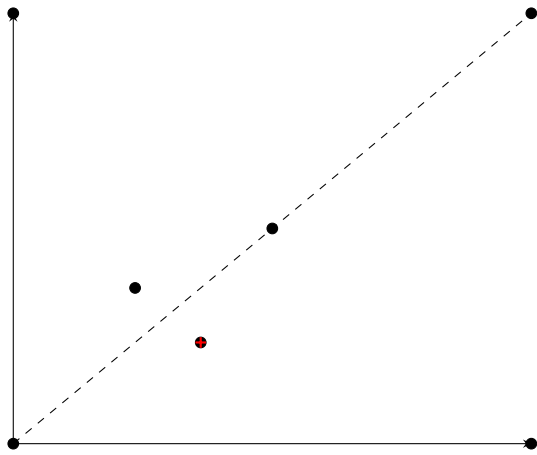
- (2, 1, 2)
 - Value: 0.015004
 - Error: 0.005564752385136
 - Relative error: 0.270540022551528
- (2, 2, 1)
 - Value: 0.021686
 - Error: 0.0000000000000001
 - Relative error: 0.0000000000000055
- (3, 1, 2)
 - Value: 0.017797
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000001
- (3, 5, 2)
 - Value: 0.056945
 - Error: 0.0000000000000007
 - Relative error: -0.0000000000000118
- (4, 3, 2)
 - Value: 0.013066
 - Error: 0.000000985435484
 - Relative error: 0.000075412735578
- (5, 4, 2)
 - Value: 0.043008
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000009
- $cu, 1$
 - Value: 0.010901
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000017
- $cu, 3$
 - Value: 0.018998
 - Error: 0.000000239104572
 - Relative error: 0.000012585722734
- $cu, 5$
 - Value: 0.019046
 - Error: 0.000000131098316
 - Relative error: 0.000006883251298



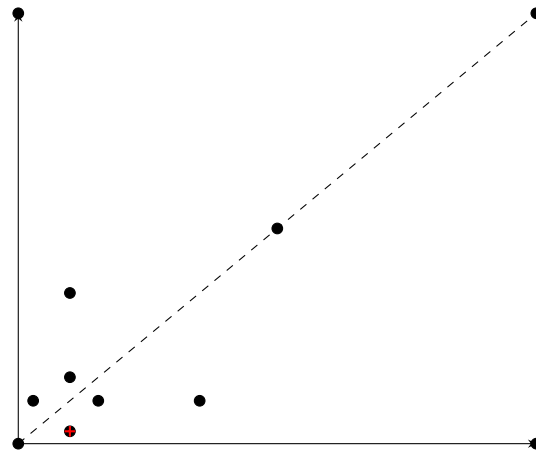
(a) RecLMTD - (2, 1, 2)



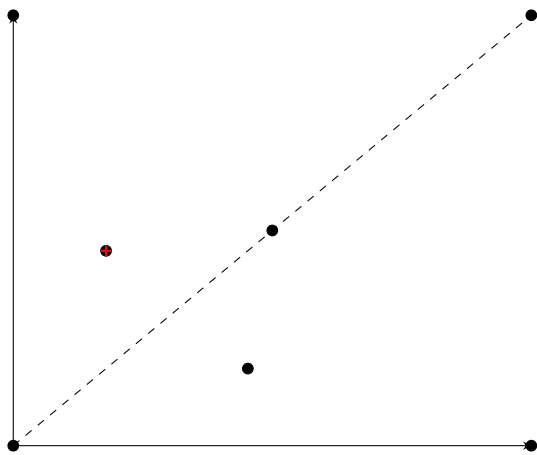
(b) RecLMTD - (2, 2, 1)



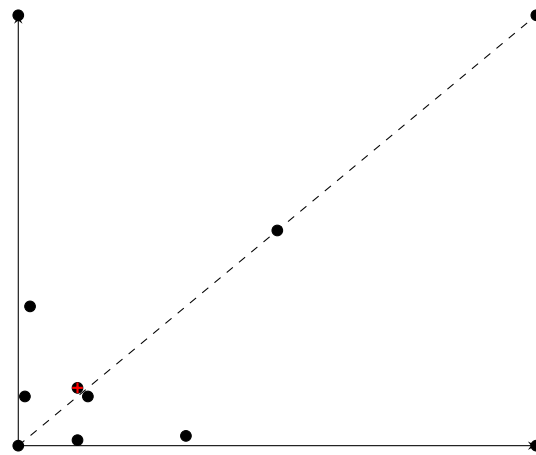
(c) RecLMTD - (3, 1, 2)



(d) RecLMTD - (3, 5, 2)



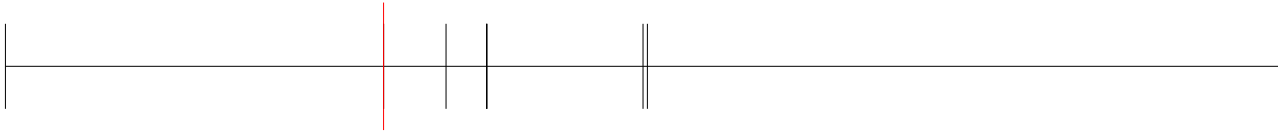
(e) RecLMTD - (4, 3, 2)



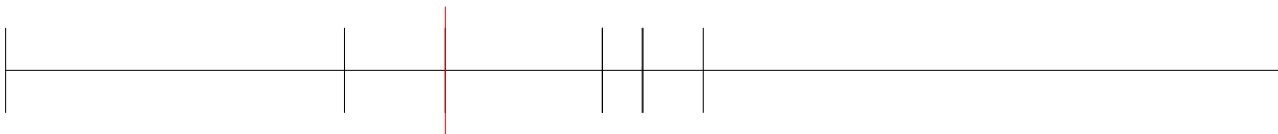
(f) RecLMTD - (5, 4, 2)



(a) RecLMTD - cu, 1



(b) *RecLMTD* - $cu, 3$



(c) *RecLMTD* - $cu, 5$

4 Balancing Breakpoints

4.1 $b^{H,\text{in}}$

- (2, 1, 2)
 - Value: 1991.1300000000538
 - Absolute error: 7.321432349272072e-11
 - Relative error: 3.6770237750785454e-14
- (2, 2, 1)
 - Value: 1449.9601170117012
 - Absolute error: 0.0
 - Relative error: 0.0
- (3, 1, 2)
 - Value: 323.02287399130614
 - Absolute error: 0.0
 - Relative error: 0.0
- (3, 5, 2)
 - Value: 2173.125654872748
 - Absolute error: 5.4569682106375694e-12
 - Relative error: 2.5111148995924616e-15
- (4, 3, 2)
 - Value: 3415.86
 - Absolute error: 0.0
 - Relative error: 0.0
- (5, 4, 2)
 - Value: 2439.3973368342085
 - Absolute error: 0.0
 - Relative error: 0.0

4.2 $b^{H,\text{out}}$

- (2, 1, 2)
 - Value: 1460.6800000000537
 - Absolute error: 5.3660187404602766e-11
 - Relative error: 3.6736442892764167e-14
- (2, 2, 1)
 - Value: 802.7501170117011
 - Absolute error: 59.14510505501141
 - Relative error: -0.06862215213722749
- (3, 1, 2)

- Value: 93.4728739913087
 - Absolute error: 0.0
 - Relative error: 0.0
- (3, 5, 2)
 - Value: 628.8356548727477
 - Absolute error: 0.0
 - Relative error: 0.0
- (4, 3, 2)
 - Value: 1876.14
 - Absolute error: 0.0
 - Relative error: 0.0
- (5, 4, 2)
 - Value: 804.5473368342084
 - Absolute error: 1.1368683772161603e-13
 - Relative error: -1.4130534341081693e-16

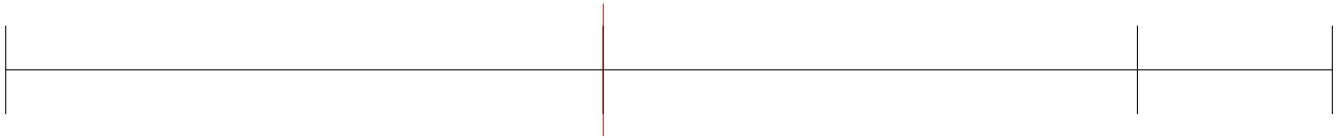
4.3 $b^{C,in}$

- (2, 1, 2)
 - Value: 318.27
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 2, 1)
 - Value: 705.16
 - Absolute error: 0.0
 - Relative error: 0.0
- (3, 1, 2)
 - Value: 137.73
 - Absolute error: 0.0
 - Relative error: 0.0
- (3, 5, 2)
 - Value: 1296.87
 - Absolute error: 0.0
 - Relative error: 0.0
- (4, 3, 2)
 - Value: 317.52
 - Absolute error: 0.0
 - Relative error: 0.0

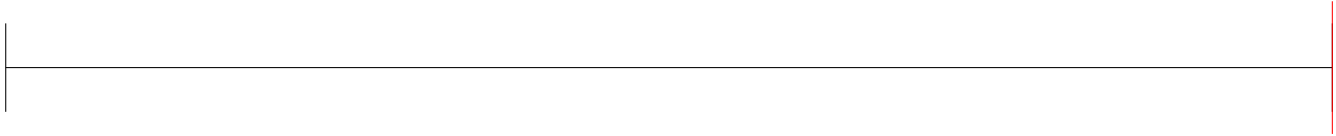
- (5, 4, 2)
 - Value: 1422.0600000000002
 - Absolute error: 0.0
 - Relative error: 0.0

4.4 $b^{C,out}$

- (2, 1, 2)
 - Value: 848.72
 - Absolute error: 48.3918096712689
 - Relative error: 0.06046495707141365
- (2, 2, 1)
 - Value: 1352.37
 - Absolute error: 0.0
 - Relative error: 0.0
- (3, 1, 2)
 - Value: 367.2799999999975
 - Absolute error: 160.21683881578775
 - Relative error: 0.7737582962584733
- (3, 5, 2)
 - Value: 2841.1600000000003
 - Absolute error: 0.0
 - Relative error: 0.0
- (4, 3, 2)
 - Value: 1857.2400000000002
 - Absolute error: 2.2737367544323206e-13
 - Relative error: 1.2242557528549464e-16
- (5, 4, 2)
 - Value: 3056.91
 - Absolute error: 0.0
 - Relative error: 0.0



(a) $t^{(H)} - (2, 2)$



(a) $t^{(H)} - (2, 1)$



(a) $t^{(H)} - (3, 2)$



(a) $t^{(H)} - (3, 2)$



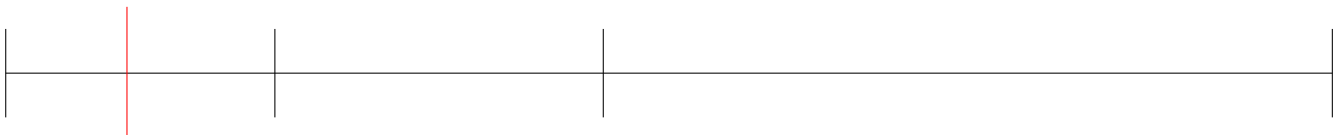
(a) $t^{(H)} - (4, 2)$



(a) $t^{(H)} - (5, 2)$



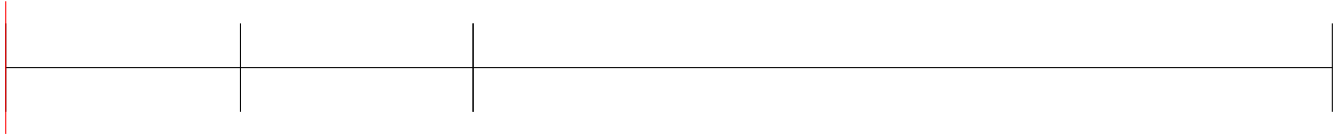
(a) $t^{(H)} - (2, 1, 2)$



(a) $t^{(H)} - (2, 2, 1)$



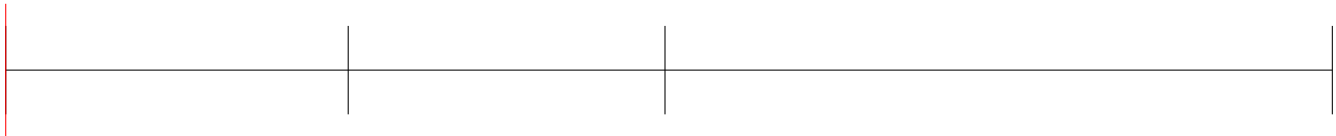
$$(a) \ t^{(H)} - (3, 1, 2)$$



$$(a) \ t^{(H)} - (3, 5, 2)$$



$$(a) \ t^{(H)} - (4, 3, 2)$$



$$(a) \ t^{(H)} - (5, 4, 2)$$



$$(a) \ t^{(C)} - (1, 3)$$



$$(a) \ t^{(C)} - (2, 2)$$



$$(a) \ t^{(C)} - (1, 3)$$



$$(a) \ t^{(C)} - (5, 3)$$



$$(a) \ t^{(C)} - (3, 3)$$



$$(a) \ t^{(C)} - (4, 3)$$



$$(a) \ t^{(C)} - (2, 1, 2)$$



$$(a) \ t^{(C)} - (2, 2, 1)$$



$$(a) \ t^{(C)} - (3, 1, 2)$$



$$(a) \ t^{(C)} - (3, 5, 2)$$



$$(a) \ t^{(C)} - (4, 3, 2)$$



$$(a) \ t^{(C)} - (5, 4, 2)$$

5 Area Breakpoints

5.1 A

- $(2, 1, 2)$
 - Value: 9.363567
 - Error: 0.000000000000613
 - Relative error: 0.000000000000065
- $(2, 2, 1)$
 - Value: 16.512573
 - Error: 0.000000000000004
 - Relative error: -0.000000000000000
- $(3, 1, 2)$
 - Value: 1.620029
 - Error: 3.186179118711760
 - Relative error: 0.662929919563720
- $(3, 5, 2)$
 - Value: 103.458296
 - Error: 0.000000000000014
 - Relative error: -0.000000000000000
- $(4, 3, 2)$
 - Value: 23.668655
 - Error: 0.000000000000000
 - Relative error: 0.000000000000000
- $(5, 4, 2)$
 - Value: 82.719688
 - Error: 0.000000000000000
 - Relative error: 0.000000000000000

5.2 A_{CU}

- $cu, 1$
 - Value: 7.527593
 - Error: 0.000000000000127
 - Relative error: -0.000000000000017
- $cu, 3$
 - Value: 13.643579
 - Error: 0.0000000000003181
 - Relative error: -0.000000000000233
- $cu, 5$

- Value: 16.235136
- Error: 0.000000000000203
- Relative error: -0.000000000000012



(a) $A^\beta - (2, 1, 2)$



(a) $A^\beta - (2, 2, 1)$



(a) $A^\beta - (3, 1, 2)$



(a) $A^\beta - (3, 5, 2)$



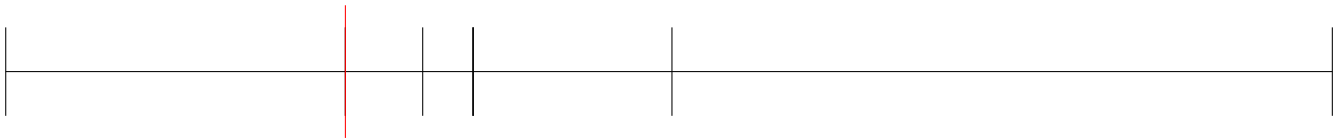
(a) $A^\beta - (4, 3, 2)$



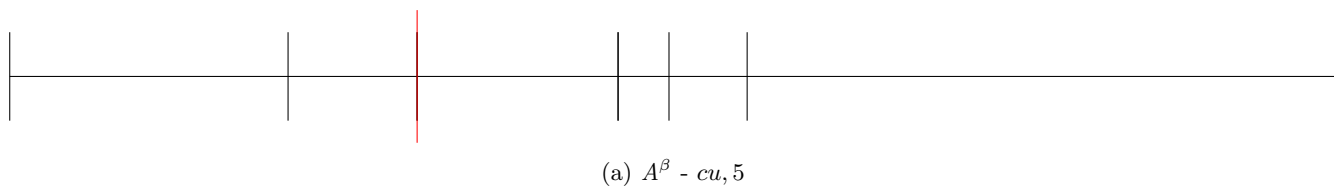
(a) $A^\beta - (5, 4, 2)$



(a) $A^\beta - cu, 1$



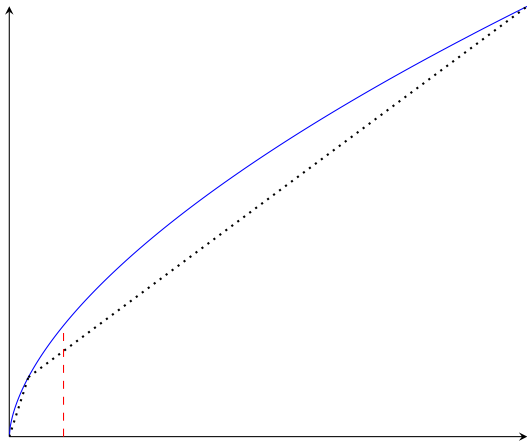
(a) $A^\beta - cu, 3$



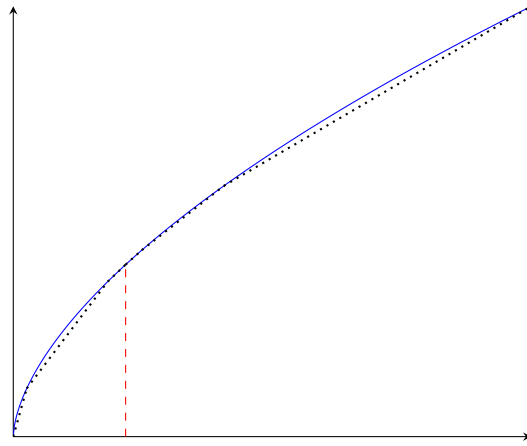
6 A^β Breakpoints

- (2, 1, 2)
 - q : 9.363567
 - $\hat{q}^\beta$: 2.947101
 - q^β : 3.827056
 - Breakpoints: [0, 3.303646498302869, 89.41176470588236]
 - Error: 0.879954334935650
 - Relative error: 0.229929843392268
- (2, 2, 1)
 - q : 16.512573
 - $\hat{q}^\beta$: 5.378843
 - q^β : 5.378843
 - Breakpoints: [0, 1.9849079917037988, 13.879417919656465, 16.512573095835464, 18.890814020083003, 30.93278282491032, 30.93278282491164, 76.14235294117645]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (3, 1, 2)
 - q : 1.620029
 - $\hat{q}^\beta$: 0.568020
 - q^β : 1.335715
 - Breakpoints: [0, 13.737212823561322, 89.41176470588236]
 - Error: 0.767694969889746
 - Relative error: 0.574744775388294
- (3, 5, 2)
 - q : 103.458296
 - $\hat{q}^\beta$: 15.525824
 - q^β : 16.175555
 - Breakpoints: [0, 8.941176470587976, 27.976182369080462, 55.82566076118323, 181.68117647058827]
 - Error: 0.649730922894840
 - Relative error: 0.040167457589017
- (4, 3, 2)
 - q : 23.668655
 - $\hat{q}^\beta$: 6.675813
 - q^β : 6.675813
 - Breakpoints: [0, 19.92063386984238, 23.668654734065058, 23.668855770059224, 181.14352941176475]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000

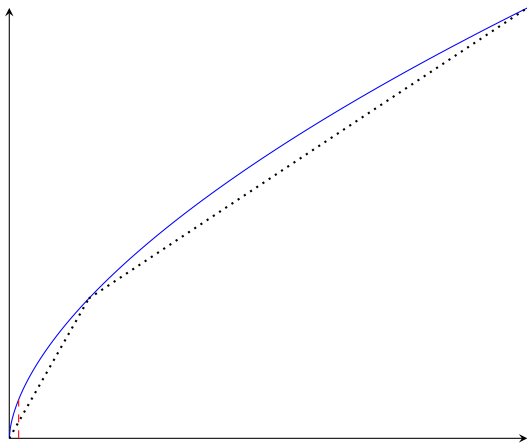
- $(5, 4, 2)$
 - q : 82.719688
 - $\hat{q}^\beta$: 13.945814
 - q^β : 14.143775
 - Breakpoints: $[0, 11.133625686778567, 51.43476571725509, 106.27900213658211, 192.3352941176471]$
 - Error: 0.197960940137149
 - Relative error: 0.013996330077818
- $cu, 1$
 - q : 7.527593
 - $\hat{q}^\beta$: 3.357327
 - q^β : 3.357327
 - Breakpoints: $[0, 7.527593053896052, 7.5275930538960525, 22.75735506822926]$
 - Error: 0.0000000000000008
 - Relative error: -0.0000000000000002
- $cu, 3$
 - q : 13.643579
 - $\hat{q}^\beta$: 4.796859
 - q^β : 4.796859
 - Breakpoints: $[0, 7.499856372317694, 10.320104112088831, 13.6435791010403, 14.376341772690381, 16.73892021814917, 16.738997089244634, 16.738997089244645, 18.852085704732907, 128.44338870924773]$
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $cu, 5$
 - q : 16.235136
 - $\hat{q}^\beta$: 5.324435
 - q^β : 5.324435
 - Breakpoints: $[0, 6.717165864481, 9.832103793393394, 14.682070969056184, 16.235135805353295, 20.94487303718885, 20.944873037188852, 21.50633939847323, 22.410514202757604, 127.1036185995295]$
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000



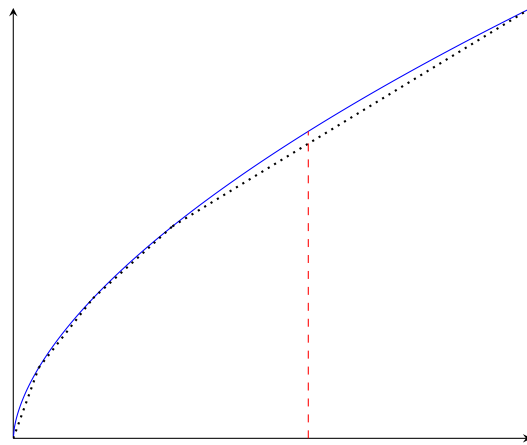
(a) $q^\beta - (2, 1, 2)$



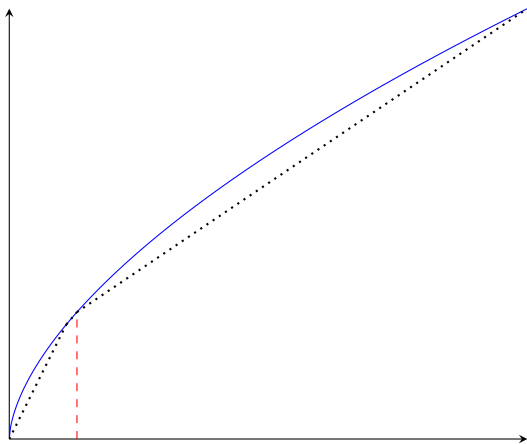
(b) $q^\beta - (2, 2, 1)$



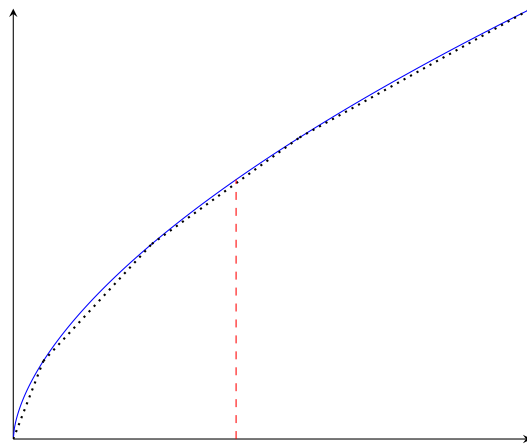
(c) $q^\beta - (3, 1, 2)$



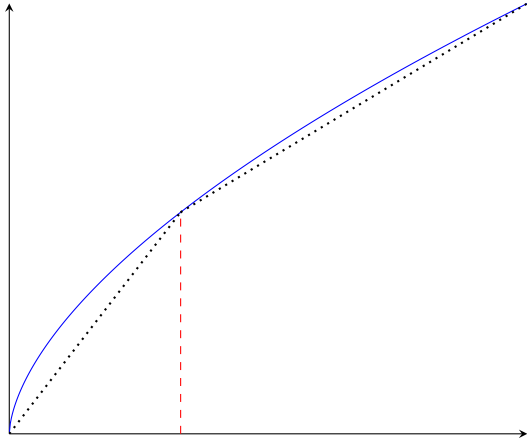
(d) $q^\beta - (3, 5, 2)$



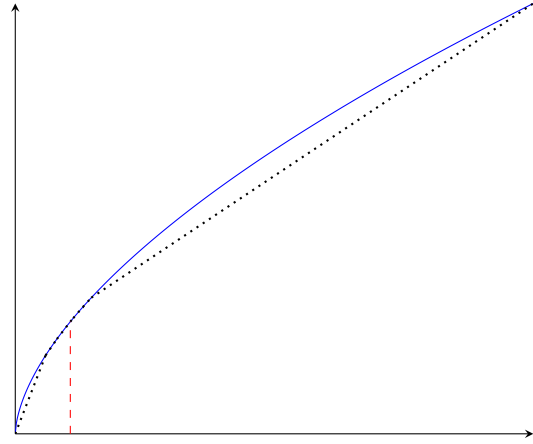
(e) $q^\beta - (4, 3, 2)$



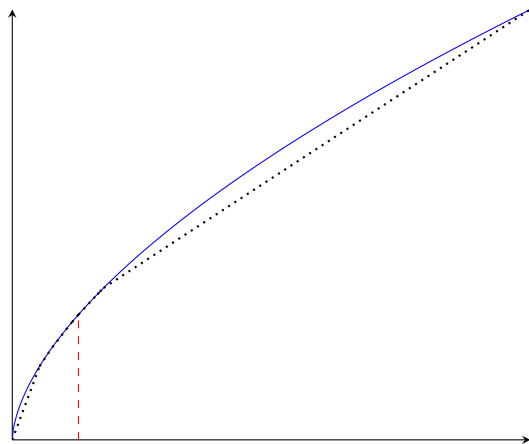
(f) $q^\beta - (5, 4, 2)$



(a) $q^\beta - cu, 1$



(b) $q^\beta - cu, 3$



(c) $q^\beta - cu, 5$

7 Other variables

Variable	Value	Variable	Value	Variable	Value
$z_{1,1,1}$	0.0	$z_{4,5,1}$	0.0	$A_{2,4,2}^\beta$	0.0
$z_{1,1,2}$	0.0	$z_{4,5,2}$	0.0	$A_{2,5,1}^\beta$	0.0
$z_{1,2,1}$	0.0	$z_{5,1,1}$	0.0	$A_{2,5,2}^\beta$	0.0
$z_{1,2,2}$	0.0	$z_{5,1,2}$	0.0	$A_{3,1,1}^\beta$	0.0
$z_{1,3,1}$	0.0	$z_{5,2,1}$	0.0	$A_{3,2,1}^\beta$	0.0
$z_{1,3,2}$	0.0	$z_{5,2,2}$	0.0	$A_{3,2,2}^\beta$	0.0
$z_{1,4,1}$	0.0	$z_{5,3,1}$	0.0	$A_{3,3,1}^\beta$	0.0
$z_{1,4,2}$	0.0	$z_{5,3,2}$	0.0	$A_{3,3,2}^\beta$	0.0
$z_{1,5,1}$	0.0	$z_{5,4,1}$	0.0	$A_{3,4,1}^\beta$	0.0
$z_{1,5,2}$	0.0	$z_{5,5,1}$	0.0	$A_{3,4,2}^\beta$	0.0
$z_{2,1,1}$	0.0	$z_{5,5,2}$	0.0	$A_{3,5,1}^\beta$	0.0
$z_{2,2,2}$	0.0	$z_{cu,2}$	0.0	$A_{4,1,1}^\beta$	0.0
$z_{2,3,1}$	0.0	$z_{cu,4}$	0.0	$A_{4,1,2}^\beta$	0.0
$z_{2,3,2}$	0.0	$z_{hu,1}$	0.0	$A_{4,2,1}^\beta$	0.0
$z_{2,4,1}$	0.0	$z_{hu,2}$	0.0	$A_{4,2,2}^\beta$	0.0
$z_{2,4,2}$	0.0	$z_{hu,3}$	0.0	$A_{4,3,1}^\beta$	0.0
$z_{2,5,1}$	0.0	$z_{hu,4}$	0.0	$A_{4,4,1}^\beta$	0.0
$z_{2,5,2}$	0.0	$z_{hu,5}$	0.0	$A_{4,4,2}^\beta$	0.0
$z_{3,1,1}$	0.0	$A_{1,1,1}^\beta$	0.0	$A_{4,5,1}^\beta$	0.0
$z_{3,2,1}$	0.0	$A_{1,1,2}^\beta$	0.0	$A_{4,5,2}^\beta$	0.0
$z_{3,2,2}$	0.0	$A_{1,2,1}^\beta$	0.0	$A_{5,1,1}^\beta$	0.0
$z_{3,3,1}$	0.0	$A_{1,2,2}^\beta$	0.0	$A_{5,1,2}^\beta$	0.0
$z_{3,3,2}$	0.0	$A_{1,3,1}^\beta$	0.0	$A_{5,2,1}^\beta$	0.0
$z_{3,4,1}$	0.0	$A_{1,3,2}^\beta$	0.0	$A_{5,2,2}^\beta$	0.0
$z_{3,4,2}$	0.0	$A_{1,4,1}^\beta$	0.0	$A_{5,3,1}^\beta$	0.0
$z_{3,5,1}$	0.0	$A_{1,4,2}^\beta$	0.0	$A_{5,3,2}^\beta$	0.0
$z_{4,1,1}$	0.0	$A_{1,5,1}^\beta$	0.0	$A_{5,4,1}^\beta$	0.0
$z_{4,1,2}$	0.0	$A_{1,5,2}^\beta$	0.0	$A_{5,5,1}^\beta$	0.0
$z_{4,2,1}$	0.0	$A_{2,1,1}^\beta$	0.0	$A_{5,5,2}^\beta$	0.0
$z_{4,2,2}$	0.0	$A_{2,2,2}^\beta$	0.0	$A_{cu,2}^\beta$	0.0
$z_{4,3,1}$	0.0	$A_{2,3,1}^\beta$	0.0	$A_{cu,4}^\beta$	0.0
$z_{4,4,1}$	0.0	$A_{2,3,2}^\beta$	0.0	$A_{hu,1}^\beta$	0.0
$z_{4,4,2}$	0.0	$A_{2,4,1}^\beta$	0.0	$A_{hu,2}^\beta$	0.0

Variable	Value	Variable	Value	Variable	Value
$A_{hu,3}^\beta$	0.0	$A_{4,3,1}$	0.0	$dt_{1,5,1}$	66.7
$A_{hu,4}^\beta$	0.0	$A_{4,4,1}$	0.0	$dt_{1,5,2}$	10.0
$A_{hu,5}^\beta$	0.0	$A_{4,4,2}$	0.0	$dt_{1,5,3}$	66.7
$A_{1,1,1}$	0.0	$A_{4,5,1}$	0.0	$dt_{2,1,1}$	111.1
$A_{1,1,2}$	0.0	$A_{4,5,2}$	0.0	$dt_{2,2,3}$	106.1
$A_{1,2,1}$	0.0	$A_{5,1,1}$	0.0	$dt_{2,3,1}$	111.1
$A_{1,2,2}$	0.0	$A_{5,1,2}$	0.0	$dt_{2,3,2}$	10.0
$A_{1,3,1}$	0.0	$A_{5,2,1}$	0.0	$dt_{2,3,3}$	183.3
$A_{1,3,2}$	0.0	$A_{5,2,2}$	0.0	$dt_{2,4,1}$	111.1
$A_{1,4,1}$	0.0	$A_{5,3,1}$	0.0	$dt_{2,4,2}$	50.042452830186846
$A_{1,4,2}$	0.0	$A_{5,3,2}$	0.0	$dt_{2,4,3}$	94.5
$A_{1,5,1}$	0.0	$A_{5,4,1}$	0.0	$dt_{2,5,1}$	111.1
$A_{1,5,2}$	0.0	$A_{5,5,1}$	0.0	$dt_{2,5,2}$	10.0
$A_{2,1,1}$	0.0	$A_{5,5,2}$	0.0	$dt_{2,5,3}$	111.1
$A_{2,2,2}$	0.0	$A_{cu,2}$	0.0	$dt_{3,1,1}$	161.1
$A_{2,3,1}$	0.0	$A_{cu,4}$	0.0	$dt_{3,2,1}$	111.1
$A_{2,3,2}$	0.0	$A_{hu,1}$	0.0	$dt_{3,2,2}$	10.0
$A_{2,4,1}$	0.0	$A_{hu,2}$	0.0	$dt_{3,2,3}$	10.0
$A_{2,4,2}$	0.0	$A_{hu,3}$	0.0	$dt_{3,3,1}$	161.1
$A_{2,5,1}$	0.0	$A_{hu,4}$	0.0	$dt_{3,3,2}$	10.0
$A_{2,5,2}$	0.0	$A_{hu,5}$	0.0	$dt_{3,3,3}$	27.8
$A_{3,1,1}$	0.0	$dt_{1,1,1}$	66.7	$dt_{3,4,1}$	144.5
$A_{3,2,1}$	0.0	$dt_{1,1,2}$	66.7	$dt_{3,4,2}$	10.0
$A_{3,2,2}$	0.0	$dt_{1,1,3}$	100.0	$dt_{3,4,3}$	10.0
$A_{3,3,1}$	0.0	$dt_{1,2,1}$	17.15006133296988	$dt_{3,5,1}$	133.4
$A_{3,3,2}$	0.0	$dt_{1,2,2}$	17.15006133296988	$dt_{4,1,1}$	122.2
$A_{3,4,1}$	0.0	$dt_{1,2,3}$	17.15006133296988	$dt_{4,1,2}$	10.0
$A_{3,4,2}$	0.0	$dt_{1,3,1}$	66.7	$dt_{4,1,3}$	100.0
$A_{3,5,1}$	0.0	$dt_{1,3,2}$	10.0	$dt_{4,2,1}$	122.2
$A_{4,1,1}$	0.0	$dt_{1,3,3}$	122.2	$dt_{4,2,2}$	10.0
$A_{4,1,2}$	0.0	$dt_{1,4,1}$	66.7	$dt_{4,2,3}$	33.3
$A_{4,2,1}$	0.0	$dt_{1,4,2}$	66.7	$dt_{4,3,1}$	122.2
$A_{4,2,2}$	0.0	$dt_{1,4,3}$	77.8	$dt_{4,4,1}$	122.2

Variable	Value	Variable	Value	Variable	Value
$dt_{4,4,2}$	122.2	$f_{1,5,1}^H$	0.0	$f_{5,4,1}^H$	0.0
$dt_{4,4,3}$	94.5	$f_{1,5,2}^H$	0.0	$f_{5,5,1}^H$	17.7
$dt_{4,5,1}$	122.2	$f_{2,1,1}^H$	0.0	$f_{5,5,2}^H$	5.435558889722429
$dt_{4,5,2}$	66.7	$f_{2,2,2}^H$	0.0	$f_{1,1,1}^C$	0.0
$dt_{4,5,3}$	111.1	$f_{2,3,1}^H$	0.0	$f_{1,1,2}^C$	0.0
$dt_{5,1,1}$	133.3	$f_{2,3,2}^H$	0.0	$f_{1,2,1}^C$	0.0
$dt_{5,1,2}$	38.9	$f_{2,4,1}^H$	0.0	$f_{1,2,2}^C$	0.0
$dt_{5,1,3}$	10.0	$f_{2,4,2}^H$	0.0	$f_{1,3,1}^C$	0.0
$dt_{5,2,1}$	74.66004781348396	$f_{2,5,1}^H$	4.774527452745195	$f_{1,3,2}^C$	0.0
$dt_{5,2,2}$	74.66004781348396	$f_{2,5,2}^H$	0.0	$f_{1,4,1}^C$	17.3
$dt_{5,2,3}$	74.66004781348391	$f_{3,1,1}^H$	0.0	$f_{1,4,2}^C$	0.0
$dt_{5,3,1}$	133.3	$f_{3,2,1}^H$	14.8	$f_{1,5,1}^C$	0.0
$dt_{5,3,2}$	53.490327808390134	$f_{3,2,2}^H$	0.0	$f_{1,5,2}^C$	0.0
$dt_{5,3,3}$	161.1	$f_{3,3,1}^H$	0.0	$f_{2,1,1}^C$	0.0
$dt_{5,4,1}$	116.7	$f_{3,3,2}^H$	0.0	$f_{2,2,2}^C$	0.0
$dt_{5,5,1}$	105.6	$f_{3,4,1}^H$	0.0	$f_{2,3,1}^C$	8.4
$dt_{5,5,2}$	105.6	$f_{3,4,2}^H$	3.789199255121066	$f_{2,3,2}^C$	0.0
$dt_{5,5,3}$	105.6	$f_{3,5,1}^H$	0.0	$f_{2,4,1}^C$	0.0
$dt_{cu,2}$	97.8	$f_{4,1,1}^H$	12.6	$f_{2,4,2}^C$	0.0
$dt_{cu,4}$	108.9	$f_{4,1,2}^H$	0.0	$f_{2,5,1}^C$	0.0
$dt_{hu,1}$	80.0	$f_{4,2,1}^H$	0.0	$f_{2,5,2}^C$	0.0
$dt_{hu,2}$	18.3	$f_{4,2,2}^H$	0.0	$f_{3,1,1}^C$	7.6
$dt_{hu,3}$	18.9	$f_{4,3,1}^H$	0.0	$f_{3,2,1}^C$	0.0
$dt_{hu,4}$	63.3	$f_{4,4,1}^H$	0.0	$f_{3,2,2}^C$	0.0
$dt_{hu,5}$	35.6	$f_{4,4,2}^H$	0.0	$f_{3,3,1}^C$	0.0
$f_{1,1,1}^H$	8.8	$f_{4,5,1}^H$	0.0	$f_{3,3,2}^C$	0.0
$f_{1,1,2}^H$	0.0	$f_{4,5,2}^H$	0.0	$f_{3,4,1}^C$	0.0
$f_{1,2,1}^H$	0.0	$f_{5,1,1}^H$	0.0	$f_{3,4,2}^C$	0.0
$f_{1,2,2}^H$	0.0	$f_{5,1,2}^H$	0.0	$f_{3,5,1}^C$	0.0
$f_{1,3,1}^H$	0.0	$f_{5,2,1}^H$	0.0	$f_{4,1,1}^C$	0.0
$f_{1,3,2}^H$	8.8	$f_{5,2,2}^H$	0.0	$f_{4,1,2}^C$	0.0
$f_{1,4,1}^H$	0.0	$f_{5,3,1}^H$	0.0	$f_{4,2,1}^C$	0.0
$f_{1,4,2}^H$	0.0	$f_{5,3,2}^H$	0.0	$f_{4,2,2}^C$	6.1

Variable	Value	Variable	Value	Variable	Value
$f_{4,3,1}^C$	0.0	$\text{RecLMTD}_{3,2,1}$	0.1	$q_{1,1,2}$	0.0
$f_{4,4,1}^C$	0.0	$\text{RecLMTD}_{3,2,2}$	0.1	$q_{1,2,1}$	0.0
$f_{4,4,2}^C$	0.0	$\text{RecLMTD}_{3,3,1}$	0.1	$q_{1,2,2}$	0.0
$f_{4,5,1}^C$	13.9	$\text{RecLMTD}_{3,3,2}$	0.026450800778310225	$q_{1,3,1}$	0.0
$f_{4,5,2}^C$	0.0	$\text{RecLMTD}_{3,4,1}$	0.1	$q_{1,3,2}$	0.0
$f_{5,1,1}^C$	0.0	$\text{RecLMTD}_{3,4,2}$	0.1	$q_{1,4,1}$	0.0
$f_{5,1,2}^C$	0.0	$\text{RecLMTD}_{3,5,1}$	0.013119967177547459	$q_{1,4,2}$	0.0
$f_{5,2,1}^C$	0.0	$\text{RecLMTD}_{4,1,1}$	0.020350921041481694	$q_{1,5,1}$	0.0
$f_{5,2,2}^C$	0.0	$\text{RecLMTD}_{4,1,2}$	0.02150212335787724	$q_{1,5,2}$	0.0
$f_{5,3,1}^C$	0.0	$\text{RecLMTD}_{4,2,1}$	0.1	$q_{2,1,1}$	0.0
$f_{5,3,2}^C$	0.0	$\text{RecLMTD}_{4,2,2}$	0.029297885186065308	$q_{2,2,2}$	0.0
$f_{5,4,1}^C$	0.0	$\text{RecLMTD}_{4,3,1}$	0.1	$q_{2,3,1}$	0.0
$f_{5,5,1}^C$	0.0	$\text{RecLMTD}_{4,4,1}$	0.02230651890829312	$q_{2,3,2}$	0.0
$f_{5,5,2}^C$	0.0	$\text{RecLMTD}_{4,4,2}$	0.024030191698075007	$q_{2,4,1}$	0.0
$\text{RecLMTD}_{1,1,1}$	0.1	$\text{RecLMTD}_{4,5,1}$	0.1	$q_{2,4,2}$	0.0
$\text{RecLMTD}_{1,1,2}$	0.1	$\text{RecLMTD}_{4,5,2}$	0.014363954130615019	$q_{2,5,1}$	0.0
$\text{RecLMTD}_{1,2,1}$	0.1	$\text{RecLMTD}_{5,1,1}$	0.1	$q_{2,5,2}$	0.0
$\text{RecLMTD}_{1,2,2}$	0.1	$\text{RecLMTD}_{5,1,2}$	0.1	$q_{3,1,1}$	0.0
$\text{RecLMTD}_{1,3,1}$	0.1	$\text{RecLMTD}_{5,2,1}$	0.1	$q_{3,2,1}$	0.0
$\text{RecLMTD}_{1,3,2}$	0.1	$\text{RecLMTD}_{5,2,2}$	0.1	$q_{3,2,2}$	0.0
$\text{RecLMTD}_{1,4,1}$	0.1	$\text{RecLMTD}_{5,3,1}$	0.1	$q_{3,3,1}$	0.0
$\text{RecLMTD}_{1,4,2}$	0.01377039538464575	$\text{RecLMTD}_{5,3,2}$	0.02884203174125638	$q_{3,3,2}$	0.0
$\text{RecLMTD}_{1,5,1}$	0.1	$\text{RecLMTD}_{5,4,1}$	0.1	$q_{3,4,1}$	0.0
$\text{RecLMTD}_{1,5,2}$	0.1	$\text{RecLMTD}_{5,5,1}$	0.1	$q_{3,4,2}$	0.0
$\text{RecLMTD}_{2,1,1}$	0.1	$\text{RecLMTD}_{5,5,2}$	0.009469696969696968	$q_{3,5,1}$	0.0
$\text{RecLMTD}_{2,2,2}$	0.1	$\text{RecLMTD}_{cu,2}$	0.009423141115769086	$q_{4,1,1}$	0.0
$\text{RecLMTD}_{2,3,1}$	0.046808918968723766	$\text{RecLMTD}_{cu,4}$	0.008535688158765427	$q_{4,1,2}$	0.0
$\text{RecLMTD}_{2,3,2}$	0.020926522199679467	$\text{RecLMTD}_{hu,1}$	0.012387051800747543	$q_{4,2,1}$	0.0
$\text{RecLMTD}_{2,4,1}$	0.1	$\text{RecLMTD}_{hu,2}$	0.045618305265862594	$q_{4,2,2}$	0.0
$\text{RecLMTD}_{2,4,2}$	0.1	$\text{RecLMTD}_{hu,3}$	0.04305097282506776	$q_{4,3,1}$	0.0
$\text{RecLMTD}_{2,5,1}$	0.1	$\text{RecLMTD}_{hu,4}$	0.015434474530129737	$q_{4,4,1}$	0.0
$\text{RecLMTD}_{2,5,2}$	0.1	$\text{RecLMTD}_{hu,5}$	0.024141568940966463	$q_{4,4,2}$	0.0
$\text{RecLMTD}_{3,1,1}$	0.1	$q_{1,1,1}$	0.0	$q_{4,5,1}$	0.0

Variable	Value	Variable	Value	Variable	Value
$q_{4,5,2}$	0.0	$t_{2,5,1}^H$	248.9	$t_{1,3,2}^C$	107.67619047619047
$q_{5,1,1}$	0.0	$t_{2,5,2}^H$	137.8	$t_{1,4,1}^C$	176.7
$q_{5,1,2}$	0.0	$t_{3,1,1}^H$	175.34864864864863	$t_{1,4,2}^C$	82.2
$q_{5,2,1}$	0.0	$t_{3,2,1}^H$	226.7	$t_{1,5,1}^C$	148.85
$q_{5,2,2}$	0.0	$t_{3,2,2}^H$	163.34633007007835	$t_{1,5,2}^C$	93.3
$q_{5,3,1}$	0.0	$t_{3,3,1}^H$	226.7	$t_{2,1,1}^C$	129.7960526315789
$q_{5,3,2}$	0.0	$t_{3,3,2}^H$	65.6	$t_{2,2,2}^C$	115.6
$q_{5,4,1}$	0.0	$t_{3,4,1}^H$	226.7	$t_{2,3,1}^C$	221.1
$q_{5,5,1}$	0.0	$t_{3,4,2}^H$	106.84594594595178	$t_{2,3,2}^C$	37.8
$q_{5,5,2}$	0.0	$t_{3,5,1}^H$	65.6	$t_{2,4,1}^C$	82.2
$q_{cu,2}$	0.0	$t_{4,1,1}^H$	271.1	$t_{2,4,2}^C$	112.86184971098264
$q_{cu,4}$	0.0	$t_{4,1,2}^H$	271.1	$t_{2,5,1}^C$	157.23125572159418
$q_{hu,1}$	0.0	$t_{4,2,1}^H$	271.1	$t_{2,5,2}^C$	93.3
$q_{hu,2}$	0.0	$t_{4,2,2}^H$	271.1	$t_{3,1,1}^C$	160.0
$q_{hu,3}$	0.0	$t_{4,3,1}^H$	271.1	$t_{3,2,1}^C$	115.6
$q_{hu,4}$	0.0	$t_{4,4,1}^H$	271.1	$t_{3,2,2}^C$	115.6
$q_{hu,5}$	0.0	$t_{4,4,2}^H$	271.1	$t_{3,3,1}^C$	111.5
$t_{1,1,1}^H$	160.0	$t_{4,5,1}^H$	271.1	$t_{3,3,2}^C$	37.8
$t_{1,1,2}^H$	133.91477272727275	$t_{4,5,2}^H$	271.1	$t_{3,4,1}^C$	176.7
$t_{1,2,1}^H$	160.0	$t_{5,1,1}^H$	198.9	$t_{3,4,2}^C$	82.2
$t_{1,2,2}^H$	160.0	$t_{5,1,2}^H$	65.60063792564067	$t_{3,5,1}^C$	166.23812949640245
$t_{1,3,1}^H$	160.0	$t_{5,2,1}^H$	198.9	$t_{4,1,1}^C$	60.0
$t_{1,3,2}^H$	160.0	$t_{5,2,2}^H$	102.51642558256174	$t_{4,1,2}^C$	60.0
$t_{1,4,1}^H$	160.0	$t_{5,3,1}^H$	198.9	$t_{4,2,1}^C$	221.7
$t_{1,4,2}^H$	160.0	$t_{5,3,2}^H$	65.60052592872505	$t_{4,2,2}^C$	115.6
$t_{1,5,1}^H$	160.0	$t_{5,4,1}^H$	126.72824858757063	$t_{4,3,1}^C$	37.8
$t_{1,5,2}^H$	160.0	$t_{5,5,1}^H$	198.9	$t_{4,4,1}^C$	176.7
$t_{2,1,1}^H$	139.290053254468	$t_{5,5,2}^H$	148.891740394162	$t_{4,4,2}^C$	82.2
$t_{2,2,2}^H$	137.8	$t_{1,1,1}^C$	90.20394736842101	$t_{4,5,1}^C$	204.4
$t_{2,3,1}^H$	248.9	$t_{1,1,2}^C$	144.10854085872572	$t_{4,5,2}^C$	93.3
$t_{2,3,2}^H$	137.8	$t_{1,2,1}^C$	115.6	$t_{5,1,1}^C$	92.62135922330097
$t_{2,4,1}^H$	198.8575471698113	$t_{1,2,2}^C$	115.6	$t_{5,1,2}^C$	90.20394736842104
$t_{2,4,2}^H$	137.8	$t_{1,3,1}^C$	111.5	$t_{5,2,1}^C$	115.6

Variable	Value	Variable	Value	Variable	Value
$t_{5,2,2}^C$	115.6	$b_{4,1,2}^{H,in}$	0.0	$b_{2,5,1}^{H,out}$	1188.3798829882792
$t_{5,3,1}^C$	111.5	$b_{4,2,1}^{H,in}$	0.0	$b_{2,5,2}^{H,out}$	0.0
$t_{5,3,2}^C$	37.8	$b_{4,2,2}^{H,in}$	0.0	$b_{3,1,1}^{H,out}$	0.0
$t_{5,4,1}^C$	156.04046242774564	$b_{4,3,1}^{H,in}$	0.0	$b_{3,2,1}^{H,out}$	3355.16
$t_{5,5,1}^C$	148.85	$b_{4,4,1}^{H,in}$	0.0	$b_{3,2,2}^{H,out}$	0.0
$t_{5,5,2}^C$	124.01079136690645	$b_{4,4,2}^{H,in}$	0.0	$b_{3,3,1}^{H,out}$	0.0
$b_{1,1,1}^{H,in}$	1408.0	$b_{4,5,1}^{H,in}$	0.0	$b_{3,3,2}^{H,out}$	0.0
$b_{1,1,2}^{H,in}$	0.0	$b_{4,5,2}^{H,in}$	0.0	$b_{3,4,1}^{H,out}$	0.0
$b_{1,2,1}^{H,in}$	0.0	$b_{5,1,1}^{H,in}$	0.0	$b_{3,4,2}^{H,out}$	859.0114711360286
$b_{1,2,2}^{H,in}$	0.0	$b_{5,1,2}^{H,in}$	0.0	$b_{3,5,1}^{H,out}$	0.0
$b_{1,3,1}^{H,in}$	0.0	$b_{5,2,1}^{H,in}$	0.0	$b_{4,1,1}^{H,out}$	3415.86
$b_{1,3,2}^{H,in}$	1408.0	$b_{5,2,2}^{H,in}$	0.0	$b_{4,1,2}^{H,out}$	0.0
$b_{1,4,1}^{H,in}$	0.0	$b_{5,3,1}^{H,in}$	0.0	$b_{4,2,1}^{H,out}$	0.0
$b_{1,4,2}^{H,in}$	0.0	$b_{5,3,2}^{H,in}$	0.0	$b_{4,2,2}^{H,out}$	0.0
$b_{1,5,1}^{H,in}$	0.0	$b_{5,4,1}^{H,in}$	0.0	$b_{4,3,1}^{H,out}$	0.0
$b_{1,5,2}^{H,in}$	0.0	$b_{5,5,1}^{H,in}$	3520.5300000000034	$b_{4,4,1}^{H,out}$	0.0
$b_{2,1,1}^{H,in}$	0.0	$b_{5,5,2}^{H,in}$	1081.1326631657912	$b_{4,4,2}^{H,out}$	0.0
$b_{2,2,2}^{H,in}$	0.0	$b_{1,1,1}^{H,out}$	1408.0	$b_{4,5,1}^{H,out}$	0.0
$b_{2,3,1}^{H,in}$	0.0	$b_{1,1,2}^{H,out}$	0.0	$b_{4,5,2}^{H,out}$	0.0
$b_{2,3,2}^{H,in}$	0.0	$b_{1,2,1}^{H,out}$	0.0	$b_{5,1,1}^{H,out}$	0.0
$b_{2,4,1}^{H,in}$	0.0	$b_{1,2,2}^{H,out}$	0.0	$b_{5,1,2}^{H,out}$	0.0
$b_{2,4,2}^{H,in}$	0.0	$b_{1,3,1}^{H,out}$	0.0	$b_{5,2,1}^{H,out}$	0.0
$b_{2,5,1}^{H,in}$	1188.3798829882792	$b_{1,3,2}^{H,out}$	1408.0	$b_{5,2,2}^{H,out}$	0.0
$b_{2,5,2}^{H,in}$	0.0	$b_{1,4,1}^{H,out}$	0.0	$b_{5,3,1}^{H,out}$	0.0
$b_{3,1,1}^{H,in}$	0.0	$b_{1,4,2}^{H,out}$	0.0	$b_{5,3,2}^{H,out}$	0.0
$b_{3,2,1}^{H,in}$	3355.16	$b_{1,5,1}^{H,out}$	0.0	$b_{5,4,1}^{H,out}$	0.0
$b_{3,2,2}^{H,in}$	0.0	$b_{1,5,2}^{H,out}$	0.0	$b_{5,5,1}^{H,out}$	3520.5300000000004
$b_{3,3,1}^{H,in}$	0.0	$b_{2,1,1}^{H,out}$	0.0	$b_{5,5,2}^{H,out}$	1081.1326631657912
$b_{3,3,2}^{H,in}$	0.0	$b_{2,2,2}^{H,out}$	0.0	$b_{1,1,1}^{C,in}$	0.0
$b_{3,4,1}^{H,in}$	0.0	$b_{2,3,1}^{H,out}$	0.0	$b_{1,1,2}^{C,in}$	0.0
$b_{3,4,2}^{H,in}$	859.0114711360286	$b_{2,3,2}^{H,out}$	0.0	$b_{1,2,1}^{C,in}$	0.0
$b_{3,5,1}^{H,in}$	0.0	$b_{2,4,1}^{H,out}$	0.0	$b_{1,2,2}^{C,in}$	0.0
$b_{4,1,1}^{H,in}$	3415.86	$b_{2,4,2}^{H,out}$	0.0	$b_{1,3,1}^{C,in}$	0.0

Variable	Value	Variable	Value	Variable	Value
$b_{1,3,2}^{C,in}$	0.0	$b_{5,2,2}^{C,in}$	0.0	$b_{4,1,2}^{C,out}$	0.0
$b_{1,4,1}^{C,in}$	3056.91	$b_{5,3,1}^{C,in}$	0.0	$b_{4,2,1}^{C,out}$	0.0
$b_{1,4,2}^{C,in}$	0.0	$b_{5,3,2}^{C,in}$	0.0	$b_{4,2,2}^{C,out}$	705.16
$b_{1,5,1}^{C,in}$	0.0	$b_{5,4,1}^{C,in}$	0.0	$b_{4,3,1}^{C,out}$	0.0
$b_{1,5,2}^{C,in}$	0.0	$b_{5,5,1}^{C,in}$	0.0	$b_{4,4,1}^{C,out}$	0.0
$b_{2,1,1}^{C,in}$	0.0	$b_{5,5,2}^{C,in}$	0.0	$b_{4,4,2}^{C,out}$	0.0
$b_{2,2,2}^{C,in}$	0.0	$b_{1,1,1}^{C,out}$	0.0	$b_{4,5,1}^{C,out}$	2841.1600000000003
$b_{2,3,1}^{C,in}$	1857.2399999999998	$b_{1,1,2}^{C,out}$	0.0	$b_{4,5,2}^{C,out}$	0.0
$b_{2,3,2}^{C,in}$	0.0	$b_{1,2,1}^{C,out}$	0.0	$b_{5,1,1}^{C,out}$	0.0
$b_{2,4,1}^{C,in}$	0.0	$b_{1,2,2}^{C,out}$	0.0	$b_{5,1,2}^{C,out}$	0.0
$b_{2,4,2}^{C,in}$	0.0	$b_{1,3,1}^{C,out}$	0.0	$b_{5,2,1}^{C,out}$	0.0
$b_{2,5,1}^{C,in}$	0.0	$b_{1,3,2}^{C,out}$	0.0	$b_{5,2,2}^{C,out}$	0.0
$b_{2,5,2}^{C,in}$	0.0	$b_{1,4,1}^{C,out}$	3056.91	$b_{5,3,1}^{C,out}$	0.0
$b_{3,1,1}^{C,in}$	1216.0	$b_{1,4,2}^{C,out}$	0.0	$b_{5,3,2}^{C,out}$	0.0
$b_{3,2,1}^{C,in}$	0.0	$b_{1,5,1}^{C,out}$	0.0	$b_{5,4,1}^{C,out}$	0.0
$b_{3,2,2}^{C,in}$	0.0	$b_{1,5,2}^{C,out}$	0.0	$b_{5,5,1}^{C,out}$	0.0
$b_{3,3,1}^{C,in}$	0.0	$b_{2,1,1}^{C,out}$	0.0	$b_{5,5,2}^{C,out}$	0.0
$b_{3,3,2}^{C,in}$	0.0	$b_{2,2,2}^{C,out}$	0.0		
$b_{3,4,1}^{C,in}$	0.0	$b_{2,3,1}^{C,out}$	1857.24		
$b_{3,4,2}^{C,in}$	0.0	$b_{2,3,2}^{C,out}$	0.0		
$b_{3,5,1}^{C,in}$	0.0	$b_{2,4,1}^{C,out}$	0.0		
$b_{4,1,1}^{C,in}$	0.0	$b_{2,4,2}^{C,out}$	0.0		
$b_{4,1,2}^{C,in}$	0.0	$b_{2,5,1}^{C,out}$	0.0		
$b_{4,2,1}^{C,in}$	0.0	$b_{2,5,2}^{C,out}$	0.0		
$b_{4,2,2}^{C,in}$	705.16	$b_{3,1,1}^{C,out}$	1216.0		
$b_{4,3,1}^{C,in}$	0.0	$b_{3,2,1}^{C,out}$	0.0		
$b_{4,4,1}^{C,in}$	0.0	$b_{3,2,2}^{C,out}$	0.0		
$b_{4,4,2}^{C,in}$	0.0	$b_{3,3,1}^{C,out}$	0.0		
$b_{4,5,1}^{C,in}$	2841.1600000000003	$b_{3,3,2}^{C,out}$	0.0		
$b_{4,5,2}^{C,in}$	0.0	$b_{3,4,1}^{C,out}$	0.0		
$b_{5,1,1}^{C,in}$	0.0	$b_{3,4,2}^{C,out}$	0.0		
$b_{5,1,2}^{C,in}$	0.0	$b_{3,5,1}^{C,out}$	0.0		
$b_{5,2,1}^{C,in}$	0.0	$b_{4,1,1}^{C,out}$	0.0		